Fusion Based Quantum Computation

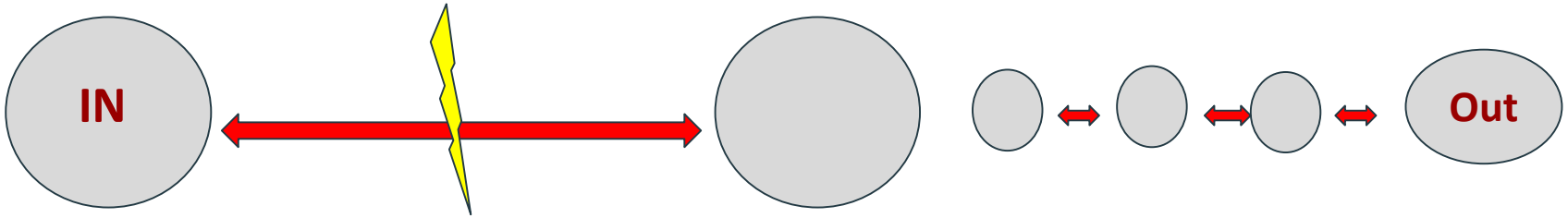
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EE5347 - Quantum Photonics And Devices

Introduction to Measurement-Based Quantum Computing

- Measurement-Based Quantum Computing (MBQC) is an alternative quantum computation model where measurements implement gate operations.
- MBQC relies on highly entangled resource states (cluster states), which are consumed through adaptive single-qubit measurements to enact quantum algorithms.
- This paradigm is also known as one-way (irreversible) or cluster-state quantum computing and is a key direction in modern quantum information science.

How can gates be implemented via Measurement



(|00>+|11>)/sqrt(2) measure 1st qubit in orthogonal basis |u><u|+|v><v|. If |0> = a|u>+b|v> and |1> = c|u>+d|v>, and the measurement return |u>, the 2nd qubit transforms to a|u>+c|1>, in other words, as if a gate was applied.

- The information held in the entanglement of qubits was converted to classical information specifying which state was measured. This can be used to correct the output state, supposing we had wanted for example, only |u> state.
- The principle of a MBQC is teleportation of information

Cluster States as a Computing Resource

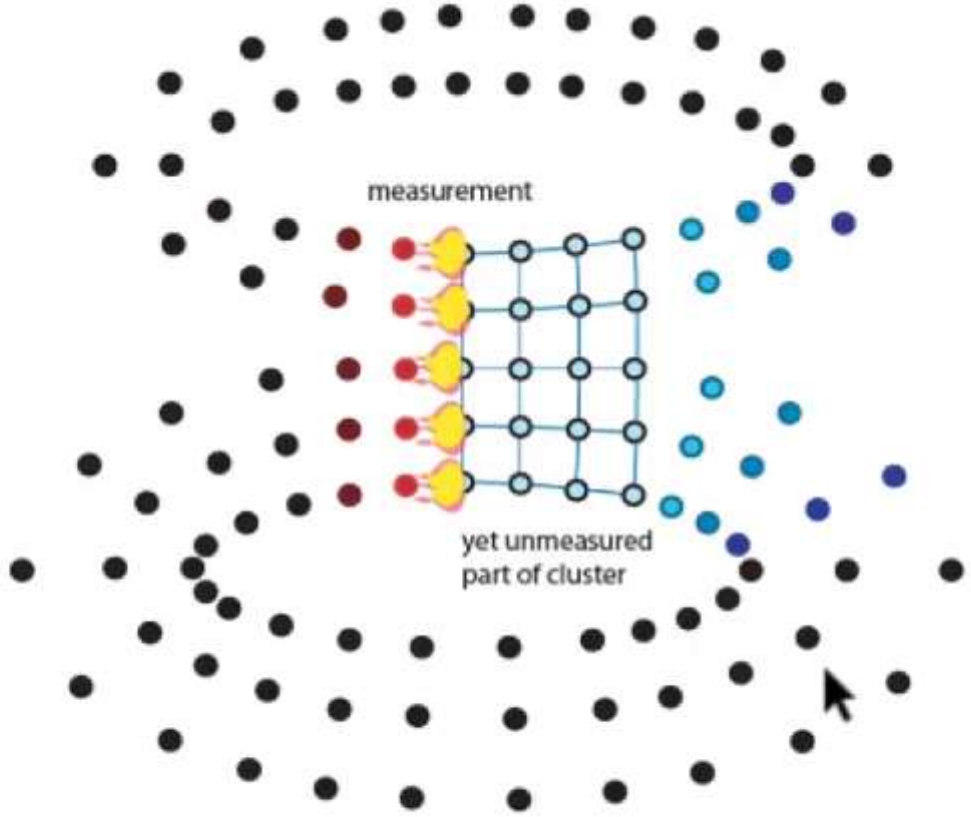
Cluster states are special types of multi-qubit entangled states, typically arranged on 2D lattices.

Preparation of cluster states is a crucial "setup" step and can be performed independently of computation.

Some of these nodes are input qubits and some are output, by applying measurements, transform and shift input qubits along some predetermines path (according to an algorithm) till it reaches the output, and a final measurement can be done. Used up states can be reused.

- Can be implemented in photonics, superconducting qubit etc
- Used up resource (measured qbits) can be refreshed in as in the picture.
- For the machine to be a universal quantum computer,

  atleast a 2d lattice is required



Conveyor belt architecture for reuse of cluster qubits.

Realizing Quantum Gates through Measurements

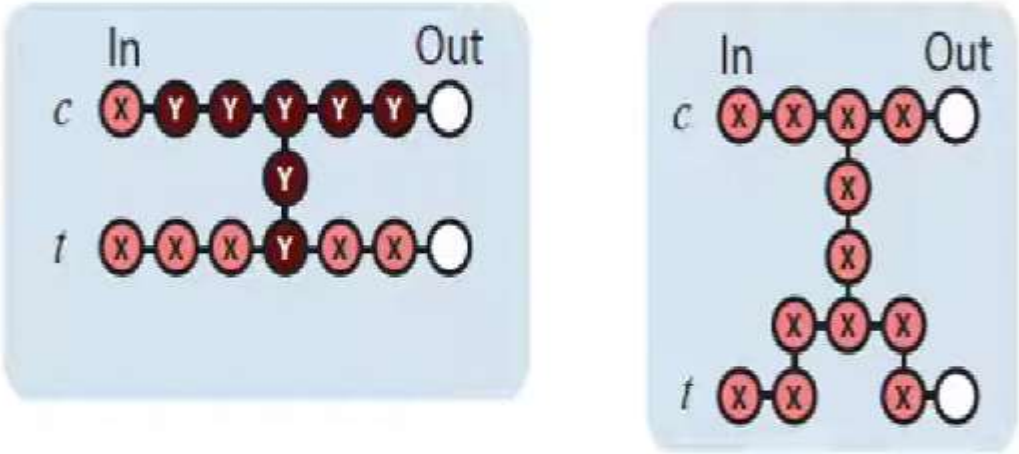
Measurement patterns on cluster states simulate quantum gates, including single-qubit rotations and two-qubit gates (e.g., CNOT).

Gate teleportation allows for non-local quantum logic by transferring quantum information across entangled nodes.

Classical processing of measurement outcomes is essential for composing more complex gate sequences.

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Two measurement patterns that realize CNOTs:



Why FBQC over MBQC

- Measurement of a photon destroys it.
- MBQC requires a large cluster state which is difficult to create error free, instead, if we can start with a large set of smaller, modular “entangled resource states”, that is much more practical.
- “Fusion” refers to a local measurement in a basis, typically bell basis, that entangles 2 photons of different resource states. This allows us to stitch together a network dynamically, and if there are errors or failures in measurement, that can be corrected by changing the measurements done using an algorithm run by a classical processor.
- Simulations show it can tolerate 10.4% loss per fusion.

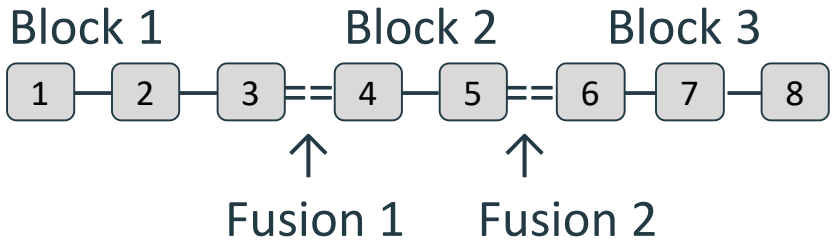
Fusion Measurement

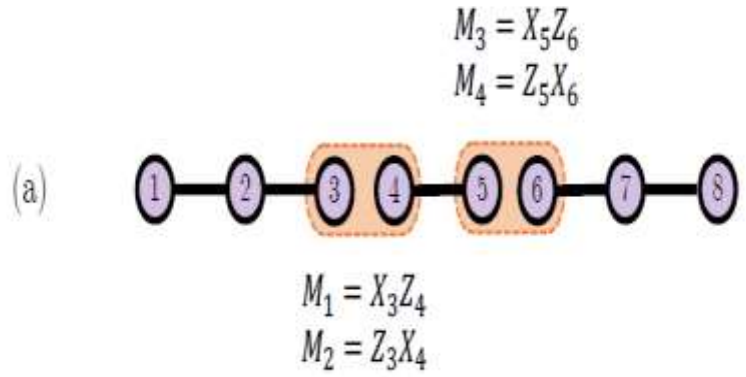
Block 1: 1 — 2 — 3

Block 2: 4 — 5

Block 3: 6 — 7 — 8

Now, we connect these blocks using *fusion measurements*.Each joins one qubit from one block with one from the next block.



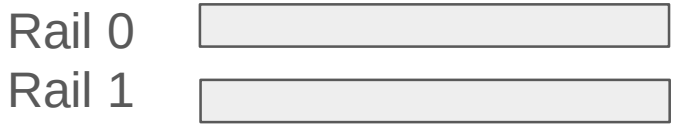




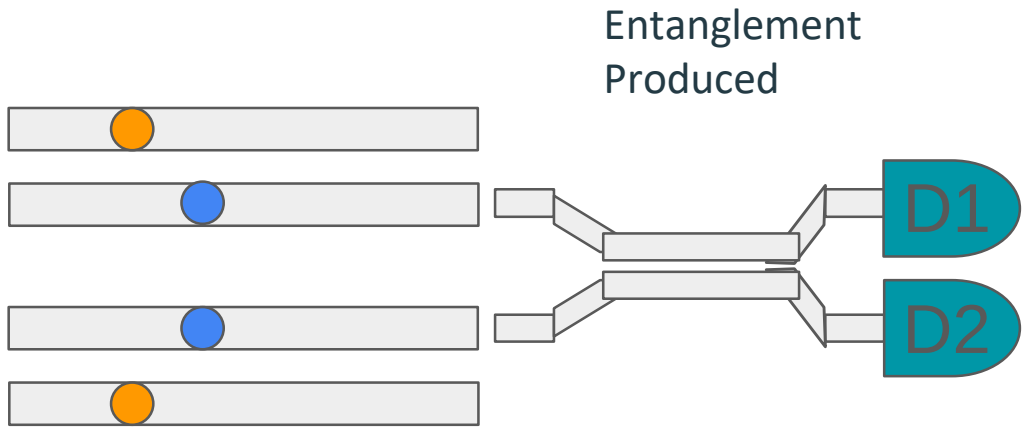
- Fusion 1: connects qubits 3 and 4 (measure X_3Z_4 and Z_3X_4)*
- Fusion 2: connects qubits 5 and 6 (measure X_5Z_6 and Z_5X_6)
- Each fusion gives two measurement results (m_1, m_2, m_3, m_4 = ±1).After these fusions, qubits 3, 4, 5, and 6 are measured and removed — they're no longer part of the final quantum state.
- The result is a 4-qubit linear graph state:
- (1)—(2)—(7)—(8) This allows us to stitch together large entanglements

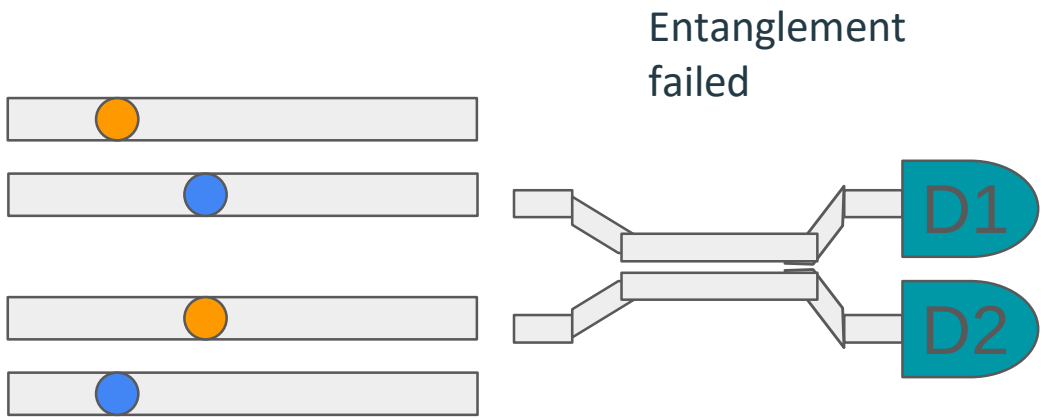
Fusion Hardware

Dual Rail Qbit



Qbit, 1 photon in 1st mode of rail 0 and 1 as states





Resource States for Fusion Networks

Small Entangled States: Fixed-size, highly entangled resource states like 4-qubit GHZ states or 6-qubit rings serve as the fundamental building blocks.

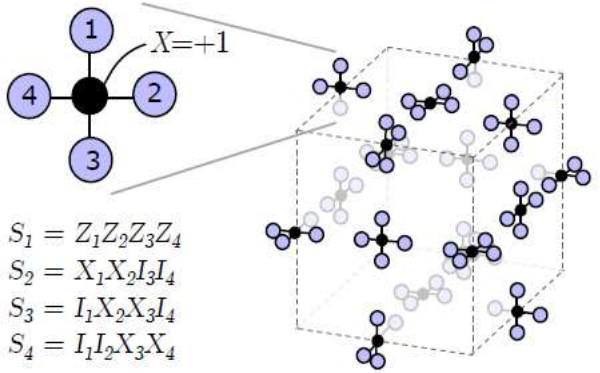
Stabilizer States: Resource states are constructed as stabilizer states, allowing straightforward representation and fusion in the Pauli group formalism.

Modular and Reproducible: Resource states are generated on-demand in constant-depth circuits, supporting modular hardware and scalability.

Encoded Resource States: Some protocols use encoded states with internal error correction (e.g., repetition codes) at the resource state level to enhance robustness.

4-Star Fusion Network

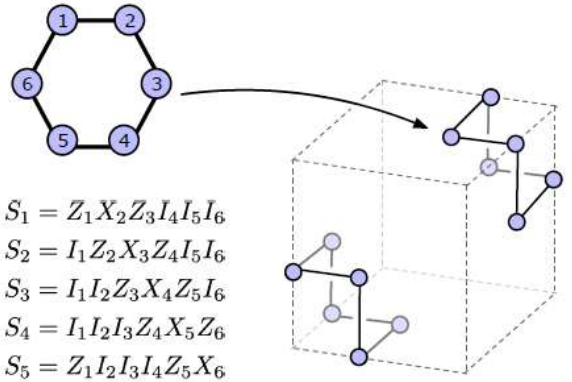
a) 4-star resource state unit cell



$S_1 = Z_1Z_2Z_3Z_4$
$S_2 = X_1X_2I_3I_4$
$S_3 = I_1X_2X_3I_4$
$S_4 = I_1I_2X_3X_4$

6-ring Fusion Network

a) 6-ring resource state unit cell



$S_1 = Z_1X_2Z_3I_4I_5I_6$
$S_2 = I_1Z_2X_3Z_4I_5I_6$
$S_3 = I_1I_2Z_3X_4Z_5I_6$
$S_4 = I_1I_2I_3Z_4X_5Z_6$
$S_5 = Z_1I_2I_3I_4Z_5X_6$
$S_6 = X_1Z_2I_3I_4I_5Z_6$

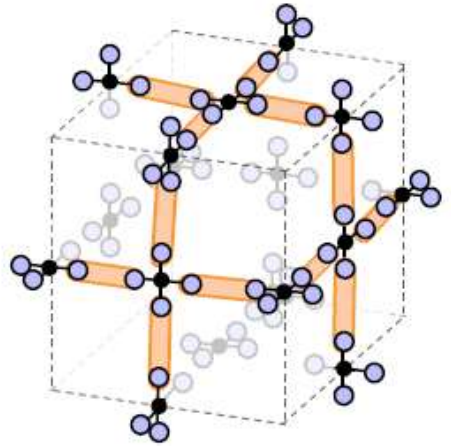
Fusion Networks — Definition & Construction

Fusion networks are formed by connecting small entangled resource states through fusion measurements, creating the computational fabric of FBQC.

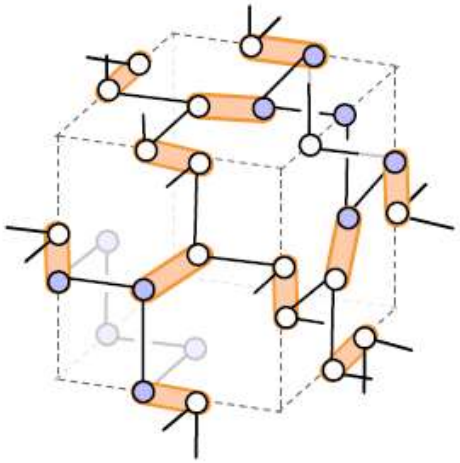
Network geometry encodes logical qubits and fault tolerance via topological error-correcting codes, promoting scalable and robust quantum computation.

Stabilizer formalism rigorously tracks how fusion measurements map to logical operations and error correction, enabling flexible network design.

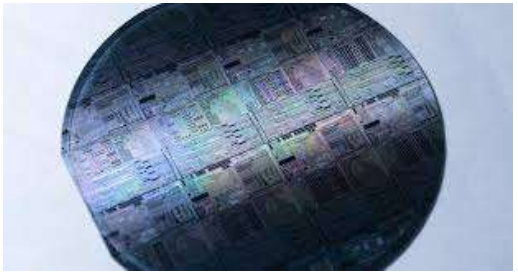
b) Fusion network



b) 6-ring fusion network

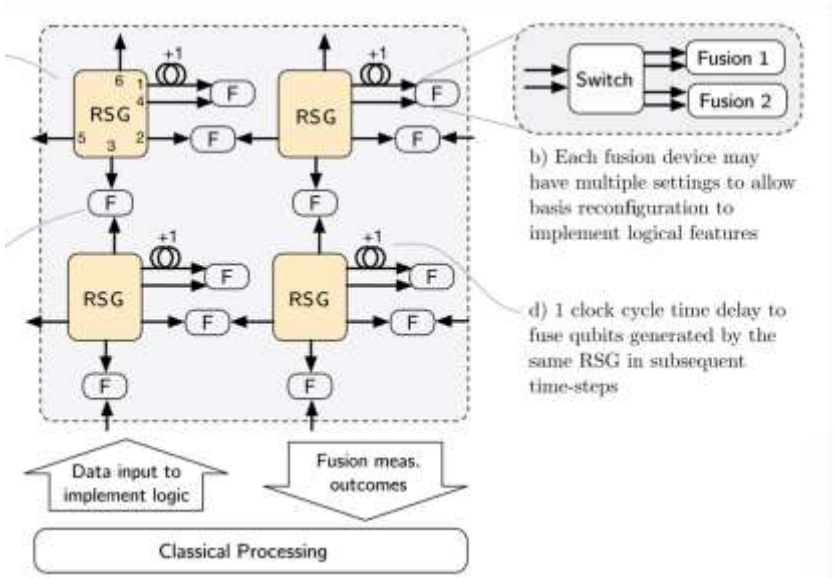


Full fusion computer



Omega Chip

A few cluster states are there in each chip, many chips are wired together, at ~ 2K for SNSPD detector to work, along with a classical processor to control the fusion measurements.

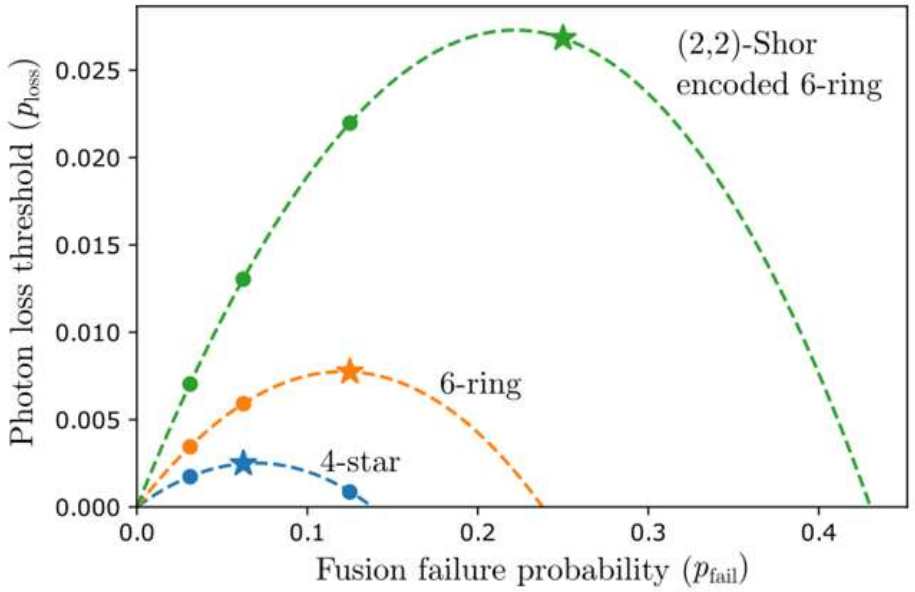


Fusion failures as well as erasures are handled by modifying the fusions performed, according to the classical algorithm.

PsiQuantum is building a FBQC



Error Model



We can see that the simpler 4 ring is not as good as 6-ring , and replacing the qubits with shor encoded qubits further improves error resilience.

The researchers then performs simulation of various resource states to determine which cluster states are good. This graph represents photon loss threshold vs failure probability, where failure probability can be affected by changing number of ancillary qbits, a technique known as boosting.

Fusion Networks and Logical Gate Implementation in FBQC

Fusion networks dynamically build large quantum codes by connecting small stabilizer-defined resource states through fusion measurements, encoding logical qubits and error correction.

Fusion operations correspond to multi-qubit Pauli measurements merging distinct modules; these fusions define stabilizers that control logical degrees of freedom and error syndromes.

Logical gates are implemented by choosing fusion measurement bases and feed-forward corrections, enabling teleportation-based gate protocols within the fusion network.

FBQC naturally supports universal quantum computation through suitable fusion network design and adaptive measurement strategies, integrating fault tolerance and scalability.